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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	**HTL**
Project / Solution Name	**Quora Chat**
Name (Leader)	**Tng Kar Ming**
Email & Phone (Leader)	**karming0425@gmail.com / +60173553022**
University Affiliation(s)	**Asia Pacific University of Technology & Innovation**
Live Prototype URL	https://quora-chat.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/jasonwong1025/quora-chat

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Quora Chat is a web-based AI-assisted quotation automation system designed as a reusable module for the Orbyte SME ecosystem.

The application provides a WhatsApp-style customer interface and a company-facing workspace. Customer enquiries are processed through a generative AI service that identifies customer intentions, asks clarifying questions and retrieves relevant public business knowledge.

Extracted requirements are validated and stored in PostgreSQL, then displayed in the Quotation Inbox with conversation evidence for employee review. The quotation workflow uses deterministic pricing rules or staff-provided estimates. Employees review and approve quotations before the system generates PDF documents and customer eSign links.

Following customer acceptance, a background worker creates an operational job and prepares a structured handover containing the confirmed requirements, agreed scope and customer commitments. Internal Team Chat allows authorized participants to collaborate using relevant business knowledge and agreement records.

The application uses TypeScript business logic for pricing, validation, approval controls and workflow transitions. Replit hosts the deployed system and provides managed OpenAI API integration.

All customer and business records are synthetic. WhatsApp, Orbyte services, email notifications and payments are simulated for the hackathon demonstration.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Typescript 6.x	Primary application language and business logic	Apache-2.0
React 19.x	Interactive user interface	MIT
TanStack Start 1.x	Full-stack application and routing	MIT
Hono 4.x	Backend HTTP Service	MIT
oRPC 1.x	Type-safe client-server communication	MIT
Drizzle ORM 0.45x	Database schema and queries	Apache-2.0
PostgreSQL 16	Relational data persistence	PostgreSQl License
Tailwind CSS 4.x	Responsive interface styling	MIT
shadcn/ui	Reusable UI components	MIT
Zod 4.x	Structured AI-output and input validation	MIT
React PDF 4.x	Quotation and billing PDF Generation	MIT
OpenAI JavaScript SDK 7.x	Calling supported AI model APIs	Apache-2.0
Node.js 22	Server-side JavaScript runtime	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI API via Replit AI Integrations	Multilingual conversations, requirement extraction, summarization, and AI-generated responses in the deployed application.	Replit-managed; usage-based
OpenRouter API	OpenAI-compatible model access during local development and testing	Usage-based; team account

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Reserved VM Deployment	Hosts the web application and background workflow worker	Reserved VM
Replit PostgreSQL Database	Stores customers, requirements, quotations, approvals, jobs, messages, and audit records	Replit-managed PostgreSQL
Replit AI Integrations	Provides managed API access to the AI model	Usage-based
GitHub	Source code management and version history	GitHub Pro (Student) / Cloud-hosted
OpenRouter	Alternative AI inference provider for local development	Usage-based

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GPT-5 mini	OpenAI via Replit AI Integrations	Pre-trained model; conversational AI, structured requirement extraction, summarization, drafting	OpenAI API service terms
GPT-5 mini	OpenAI via OpenRouter	Pre-trained model; local development and testing	OpenRouter and model-provider terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic SME Business Dataset	Team-created; project repository	Synthetic	Team-owned	N
Synthetic Customer Conversation Scenarios	Team-created; project repository	Synthetic	Team-owned	N
Product Catalogues and Pricing Rules	Team-created; project repository	Synthetic	Team-owned	N
Business Knowledge and SOP Examples	Team-created; project repository	Synthetic	Team-owned	N
Workflow and Evaluation Test Cases	Team-created; project repository	Synthetic	Team-owned	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	AI-assisted implementation, application configuration, debugging, and deployment support	Major
Claude / Claude Code	Code implementation, refactoring, debugging, and technical improvements	Major
OpenAI Codex	Code implementation, code review, testing, and debugging	Major
Kiro	AI-assisted implementation and development tasks	Moderate
/brag by Latent Spaces	AI-assisted promotional video planning, visual storytelling and video generation for the Quora Chat presentation	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used Replit Agent, Claude Code, Codex and Kiro for implementation, debugging and testing. We also used the open source /brag agent skill to assist with presentation video production. Team members defined the project requirements, reviewed generated outputs and validated the application using TypeScript, Biome, unit tests and database integration tests.

As of 24 October 2026, 63 unit tests and 36 integration tests passed. All AI-assisted code, designs and presentation materials were reviewed and adapted by the team, which remains responsible for their accuracy, originality and quality.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core	Application development, project configuration, and AI-assisted development
Replit	Reserved VM Hosting	Hosting the deployed prototype and persistent background worker
Replit	Replit AI Integration	Accessing the OpenAI model for application inference

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the cloud deployment environment and managed AI integration used by the Quora Chat prototype. The application runs on a Replit Reserved VM, stores workflow records in PostgreSQL and accesses OpenAI models through Replit AI Integrations. Replit Agent also supported development and debugging.

These services are important to the current implementation and simplify deployment during the hackathon. However, the application is designed around portable web technologies and could be migrated to alternative hosting, database and AI providers.

RunPod was not used because the prototype relies on hosted pre-trained model APIs rather than GPU-based training or inference.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Quora Chat is a hackathon prototype containing synthetic customer identities, employee profiles, business catalogues, conversations, requirements, quotations, invoices and operational records. No real customer or sensitive business records are intentionally used.

The system stores conversation history, requirement versions, approvals, jobs and activity records in PostgreSQL. Relevant customer messages and permitted business knowledge may be transmitted to the configured third-party AI provider for inference.

API credentials are configured through environment variables rather than intentionally included in source code. Public and internal business knowledge are separated through application-level retrieval rules.

The current prototype uses simulated staff identities and does not implement production-grade authentication or authorization. Accordingly, it is intended only for synthetic demonstration data.

Records remain in the demonstration database until they are manually reset or deleted. Real-world deployment would require authenticated access, server-enforced permissions, defined retention policies and appropriate compliance with Malaysia's Personal Data Protection Act.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Generative AI may misunderstand multilingual messages, omit customer requirements, produce inaccurate summaries or generate unsupported responses. The prototype mitigates these risks through structured output validation, configurable requirement questions, evidence references, clarification prompts and human intervention.

AI does not independently establish final quotation prices or authorize commercial commitments. Deterministic business logic handles pricing, requirement versions, approval conditions and workflow transitions.

External AI APIs introduce latency, rate-limit, availability and usage-cost risks. Background workers include retry mechanisms, and a scripted replay mode supports reproducible demonstrations when live AI is unavailable.

The prototype uses simulated staff identities instead of authenticated sessions. Application-level access checks therefore do not provide production-grade authorization, and document endpoints require additional protection before real deployment.

WhatsApp, Orbyte integration, email delivery and payments are simulated. Full attachment-content extraction and real-world multilingual accuracy evaluation remain outside the current prototype scope.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Tng Kar Ming

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24 October 2026